5.9.2 The Theory of Plate Tectonics

After completing this chapter you should be able to relate the development of the Universe and the dynamic structure of the Earth to models, theories and laws and the influence of time.

For Questions 1 to 19 choose the best answer (A), (B), (C) or (D).

1. Which of the following gives evidence for the theory of plate tectonics?

(A) Mountain ranges occur in the same regions as plate boundaries.
(B) Earthquakes do not occur in mountainous regions.
(C) Most of the world's volcanoes circle the Pacific Ocean.
(D) Volcanic action releases molten material onto the surface of the Earth.

2. The diagram shows how Africa and South America seem to fit against each other.



Which scientific theory best accounts for this 'fit'?

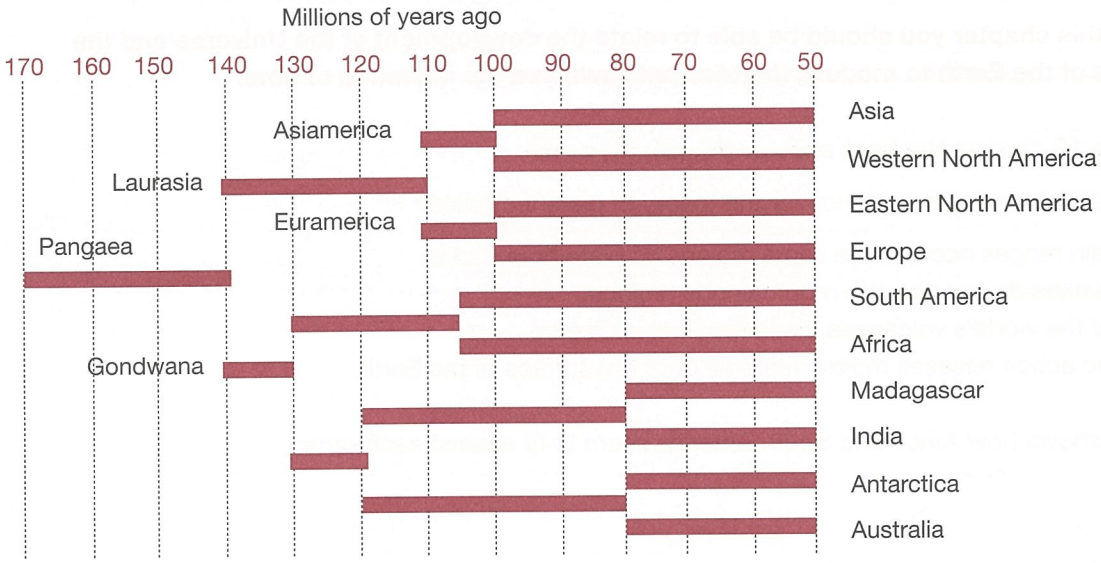
(A) The big bang theory.
(B) The theory of evolution.
(C) The theory of natural selection.
(D) The theory of plate tectonics.

3. Which statement best accounts for earthquakes?

(A) Rocks in the crust sink into rocks in the asthenosphere.
(B) The Moon and Sun place gravitational forces on the Earth that cause its crust to move.
(C) The Earth's crust is made of large blocks that are continually pushing and moving against each other.
(D) Chemical reactions in molten magna produce uplifting forces on the crust.

Use the following information to answer the next TWO questions.

Separation of land masses



4. Which land masses were formed from Euramerica?

(A) Asia and Western North America.
(B) Western North America and Eastern North America.
(C) Eastern North America and Europe.
(D) Europe and South America.

5. How long has Africa existed as a separate land mass?

(A) 105 million years.
(B) 115 million years.
(C) 130 million years.
(D) 170 million years.

6. According to this information, what name was given to the first land mass on Earth?

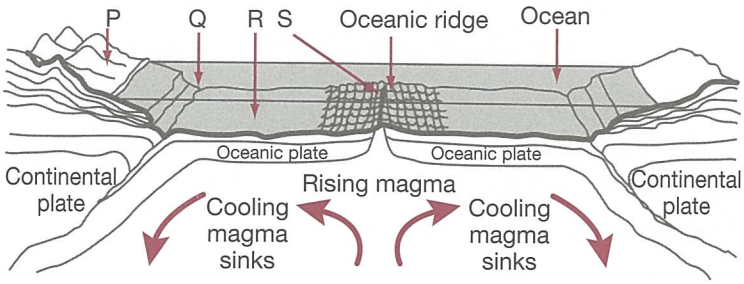
(A) Gondwanaland.
(B) Gondwana.
(C) Laurasia.
(D) Pangaea.

7. Which of the following *does not* support the theory of plate tectonics?

(A) Matching flora and fauna types in different continents.
(B) Correlated rock sequences in different countries.
(C) The distribution of sedimentary rocks throughout the world.
(D) The distribution of earthquakes throughout the world.

Use the following information to answer the next FOUR questions.

The diagram represents geological activity at plate boundaries.



8. What can be inferred from this diagram?

(A) New crust is forming at ocean ridges.
(B) The rising magma pushes the oceanic plates together.
(C) The continental plates are expanding under the oceanic plates.
(D) The movement of the continental plates towards each other forms the ocean ridge.

9. Which region will have the most volcanic activity?

(A) P
(B) Q
(C) R
(D) S

10. How does the age of the rocks near the oceanic ridge compare with those further from it?

(A) They are older.
(B) They are younger.
(C) They are about the same age.
(D) Some will be older, some will be younger.

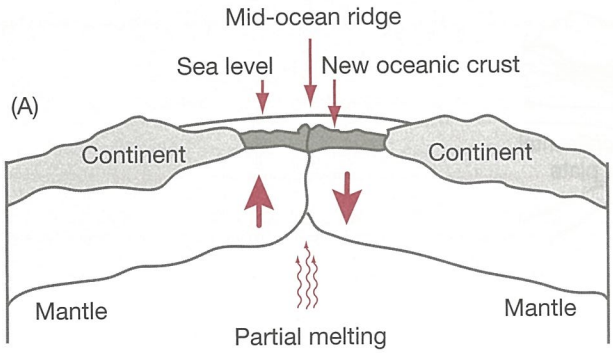
11. What name is given to the layer of the Earth in which the magma convection currents are occurring?

(A) The asthenosphere.
(B) The crust.
(C) The mantle.
(D) The outer core.

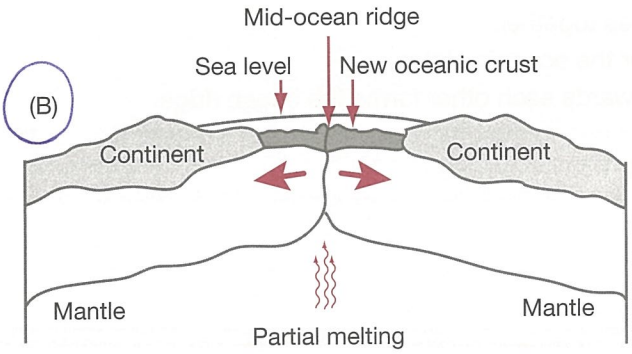
12. Crustal movement associated with the formation of mid-ocean ridges occurs because of convection currents under the crustal blocks.

Which diagram shows the correct direction of movement of the blocks either side of the mid-ocean ridge?

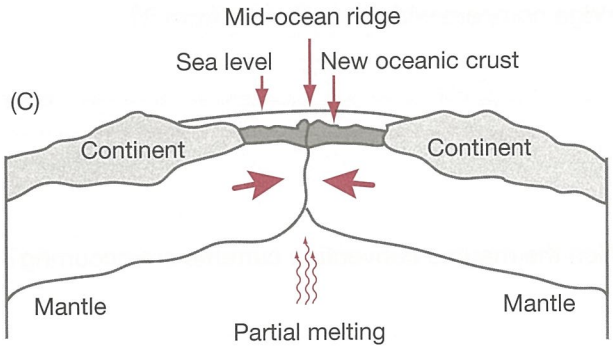
(A)



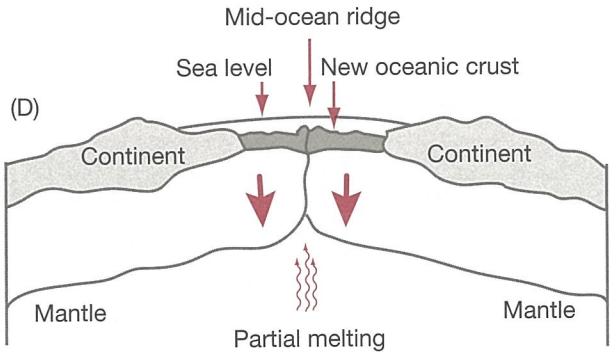
(B)



(C)



(D)

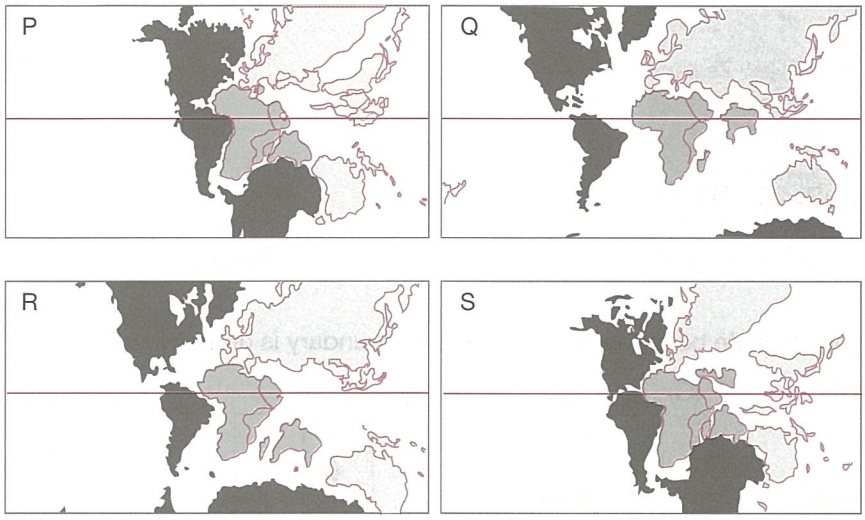


13. What is *not* a source of the heat energy that drives convection currents in the mantle that cause plate tectonics?

(A) The Earth's core.
(B) Radioactivity within the Earth.
(C) The mantle of the Earth.
(D) Friction between the crustal plates.

Use the following information to answer the next TWO questions.

The diagrams show proposed positions of the continents on Earth at various stages between 50 and 200 million years ago.



14. What is the correct sequence of these diagrams from earliest to latest?

(A) Q, R, P, S.
(B) P, R, S, Q.
(C) P, S, R, Q.
(D) S, P, R, Q.

15. What is thought to have caused this movement of the land masses?

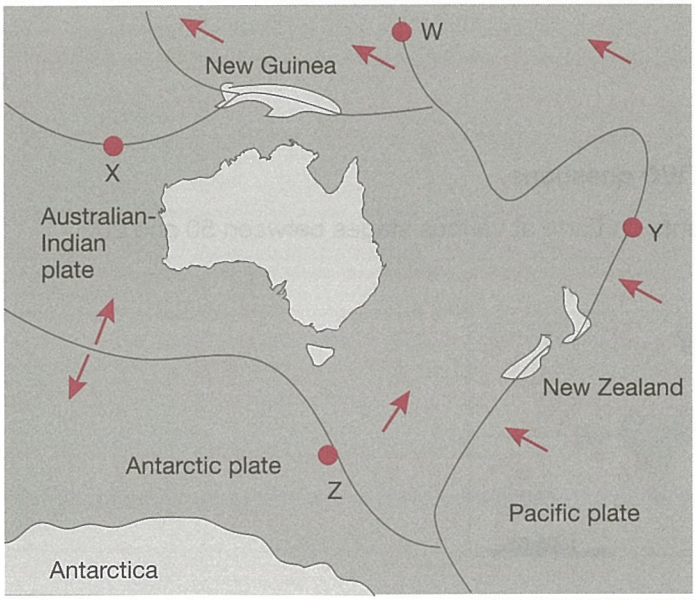
(A) Convection currents in magma in the asthenosphere.
(B) Expansion forces in the crust of the Earth as it cooled down.
(C) Movement of crustal blocks away from each other.
(D) The theory of plate tectonics.

16. Which of the following best supports the idea of continental drift?

(A) The Grand Canyon in America.
(B) The wide distribution of oil across the globe.
(C) The discovery of coal in Antarctica.
(D) The extinction of the dinosaurs.

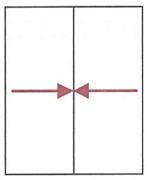
Use the following information to answer the next THREE questions.

The map shows the movement of crustal plates around Australia.



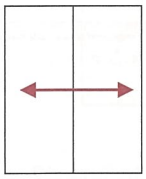
17. The diagrams show four different types of plate boundaries. What type of boundary is under New Guinea?

(A)



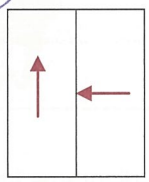
Convergent

(B)



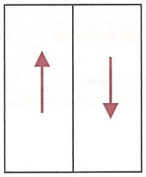
Divergent

(C)



Subduction

(D)



Transform

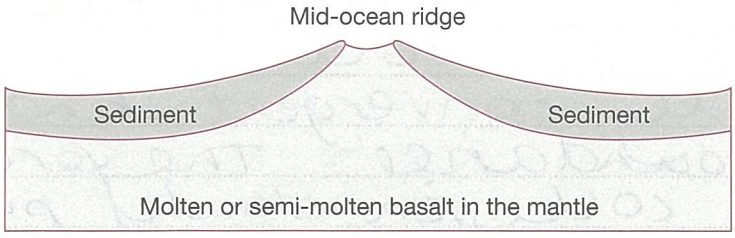
18. Where would we be most likely to find volcanoes?

(A) Antarctica and Australia.
(B) Australia and New Guinea.
(C) Antarctica and New Zealand.
(D) New Guinea and New Zealand.

19. Given that mid-ocean ridges form at divergent boundaries, where are we most likely to find a mid-ocean ridge?

(A) W
(B) X
(C) Y
(D) Z

20. The diagram shows a mid-ocean ridge.



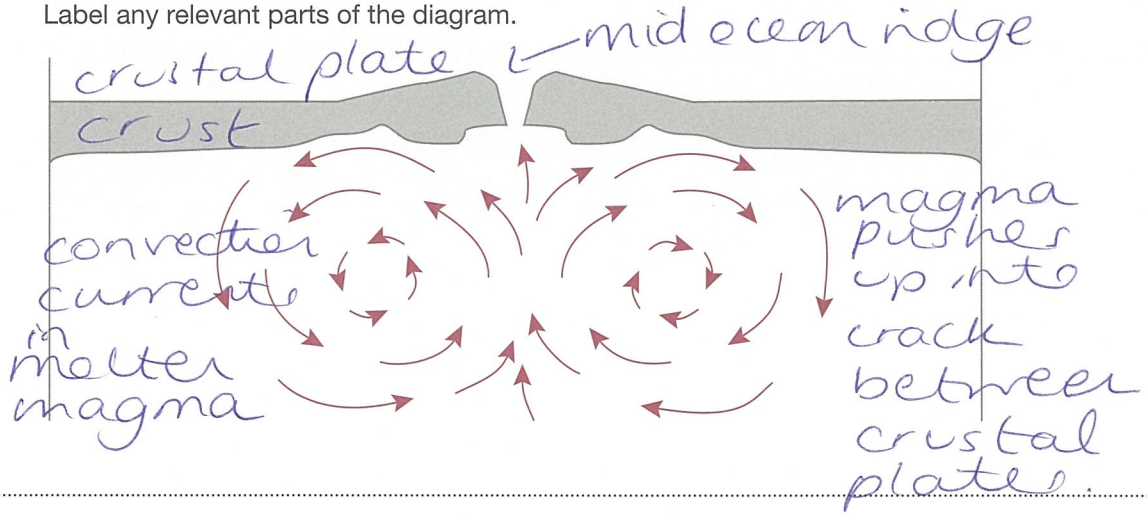
(a) Outline how mid-ocean ridges form.

Convection currents in the mantle under the ocean push upwards in the region in which the ridge forms, bending and eventually cracking the crust allowing molten magma to flow into the ocean. It cools and solidifies and builds up to form the mid-ocean ridge.

(b) Suggest a reason for the thicker covering of sediment further away from the mid-ocean ridge.

The sediment is older and had more time for additional layers to build up.

21. The diagram represents an aspect of plate tectonics. Explain what the diagram is showing. Label any relevant parts of the diagram.



Molten magma pushes up into the cracks between the crustal plates and forms underwater volcanoes. The magma cools, solidifies and slowly builds up to form the mid-ocean ridge.

22. Outline how the theory of plate tectonics explains the formation of mountain chains.

Mountain chains form along the boundaries between colliding crustal plates especially convergent and subduction boundaries. The forces involved in the collision slowly pushes the edges of the plates upwards to form mountains.